

Linear Algebra – Problem Set 2 — Solutions

Row Echelon Form & the Determinant · Full solutions · Original problems in the spirit of MIT 18.06 (Strang)

Problem 1

Compute $\det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 1 & 1 \end{bmatrix}$ by reducing to row echelon form. Use row replacements only — no swaps, no scalings — and state why your answer is legitimate.

Clear the first column below the pivot:

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - R_1 : \quad \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & -1 & -2 \end{bmatrix}$$

Clear the second column below its pivot:

$$R_3 \leftarrow R_3 + R_2 : \quad \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & -5 \end{bmatrix}$$

This is upper triangular, so the determinant is the product of the diagonal entries, $1 \cdot 1 \cdot (-5) = -5$.

Legitimacy: all three operations are row *replacements*. Each one leaves the determinant exactly unchanged, so the echelon matrix has the same determinant as A .

$\det A = -5$

Problem 2

Compute $\det \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ by reducing to row echelon form. A leading entry of zero forces a swap. Perform the swap, finish the reduction, and explain exactly where the swap's effect on the determinant enters your answer.

The first pivot is 0, so swap $R_1 \leftrightarrow R_2$:

$$M = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 0 \end{bmatrix}, \quad \det M = -\det A \quad (\text{one swap flips the sign}).$$

Now finish with replacements only:

$$R_3 \leftarrow R_3 - R_1 : \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & -1 \end{bmatrix}, \quad R_3 \leftarrow R_3 - R_2 : \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \end{bmatrix}.$$

Those two steps changed nothing, so $\det M = 1 \cdot 1 \cdot (-3) = -3$.

Where the sign enters: $\det M = -\det A$ means $\det A = -\det M$. Had you forgotten the swap, you would have reported -3 — the right magnitude, the wrong sign.

$\det A = 3$

Problem 3

Let $A = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 0 & 1 & 1 \end{bmatrix}$. Without computing the determinant, use row echelon form to

decide whether $\det A$ is zero, and say which single observation settles it.

Look at the rows before eliminating anything. Row 3 is exactly twice row 2:

$$2 \cdot [1 \ 2 \ 3 \ 4] = [2 \ 4 \ 6 \ 8].$$

Row replacement turns R_3 into zero: apply $R_3 \leftarrow R_3 - 2R_2$. The result has a zero row, hence zero determinant — and since that step was a replacement, $\det A$ itself is zero.

In echelon language the same fact appears as a row of zeros and only three pivots:

$$\text{REF}(A) = \begin{bmatrix} 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\det A = 0$ — **the settling observation is $R_3 = 2R_2$.**

Problem 4

A common habit is to normalize each row while reducing (dividing the pivot row by its leading entry). Explain why this is a bad idea when the goal is the determinant. Give a 2×2 example making the cost concrete, and say which single step type you should avoid.

Scaling a row by c multiplies the determinant by c and by nothing else. Once you normalize, the product of the diagonal no longer equals $\det A$ — you would have to recover the discarded factors, which defeats the purpose of the shortcut.

Take $A = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$, so $\det A = 8$. Normalize both rows and you get I_2 , whose determinant is 1. You have thrown away the factor $2 \cdot 4 = 8$ and left yourself no record of it in the matrix you are about to read off. (You could still multiply the discarded factors back in — but then the method saves nothing over cofactor expansion on a 2×2 .)

Replacements are fine, swaps are fine if you count them, so the single step type to avoid is **scaling a row**.

Avoid row scaling; a normalized reduction yields $\det(\text{normalized}) \neq \det A$.

Problem 5

Starting from $A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ 6 & 5 & 1 \end{bmatrix}$, the following operations are performed in order:

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - 3R_1, \quad R_3 \leftarrow R_3 - 2R_2,$$

producing an upper triangular matrix U . Find $\det A$ *without* redoing the elimination.

All three operations are row replacements, so $\det U = \det A$. The elimination has already been done for you; the only work left is to recognize what U is.

Apply them in order to check the shape:

$$R_2 \leftarrow R_2 - 2R_1 : [4 \ 3 \ 1] - [4 \ 2 \ 0] = [0 \ 1 \ 1]$$

$$R_3 \leftarrow R_3 - 3R_1 : [6 \ 5 \ 1] - [6 \ 3 \ 0] = [0 \ 2 \ 1]$$

$$R_3 \leftarrow R_3 - 2R_2 : [0 \ 2 \ 1] - [0 \ 2 \ 2] = [0 \ 0 \ -1]$$

So $U = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$ and $\det A = 2 \cdot 1 \cdot (-1) = -2$.

$\det A = -2$

Problem 6

Let $A = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 2 & 3 & 2 & 5 \\ 1 & 2 & 1 & 3 \end{bmatrix}$. Reduce to row echelon form. Then (a) report the rank, and (b)

exhibit the exact relation among the rows that forces $\det A = 0$.

First make the first pivot 1 (it already is) and clear column 1:

$$R_3 \leftarrow R_3 - 2R_1, \quad R_4 \leftarrow R_4 - R_1 : \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Row 3 now *equals* row 2, so one more replacement kills it: $R_3 \leftarrow R_3 - R_2 = [0 \ 0 \ 0 \ 0]$. Clearing column 2 below the pivot: $R_4 \leftarrow R_4 - R_2 = [0 \ 0 \ -1 \ 0]$.

$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

That is not yet echelon form — a zero row is only allowed at the *bottom*, and row 4 still carries an entry in column 3. Move it down with $R_3 \leftrightarrow R_4$:

$$\text{REF}(A) = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(The swap flips the sign of the determinant, which is irrelevant here since we only need to know it is zero.)

- (a) There are three nonzero rows, so the rank is 3.
- (b) The row relation, read off from the $R_3 \leftarrow R_3 - 2R_1$ step: row 3 of the original matrix was $2 \times$ row 1 plus row 2. Indeed

$$2[1 \ 1 \ 0 \ 2] + [0 \ 1 \ 2 \ 1] = [2 \ 3 \ 2 \ 5].$$

One row is a combination of the others, so the rows are linearly dependent and $\det A = 0$.

(a) $\text{rank}(A) = 3$. **(b)** $R_3 = 2R_1 + R_2$, so $\det A = 0$.

Problem 7

Let $A_t = \begin{bmatrix} t & 1 & 0 & 0 \\ 0 & t & 1 & 0 \\ 0 & 0 & t & 1 \\ 1 & 0 & 0 & t \end{bmatrix}$. Use row echelon form to find all *real* values of t for which $\det A_t = 0$. Justify that your list is complete.

For $t \neq 0$ the elimination runs cleanly. The first pivot is t , so clear the entry below it:

$$R_4 \leftarrow R_4 - \frac{1}{t}R_1 : \quad [1 \ 0 \ 0 \ t] - \frac{1}{t}[t \ 1 \ 0 \ 0] = [0 \ -\frac{1}{t} \ 0 \ t].$$

The second pivot is t again; the entry to clear is $-\frac{1}{t}$, and the multiplier is $\frac{1}{t^2}$:

$$R_4 \leftarrow R_4 + \frac{1}{t^2}R_2 : \quad [0 \ 0 \ \frac{1}{t^2} \ t].$$

Third pivot t ; the entry to clear is $\frac{1}{t^2}$, so the multiplier is $\frac{1}{t^3}$:

$$R_4 \leftarrow R_4 - \frac{1}{t^3}R_3 : \quad \begin{bmatrix} t & 1 & 0 & 0 \\ 0 & t & 1 & 0 \\ 0 & 0 & t & 1 \\ 0 & 0 & 0 & t - \frac{1}{t^3} \end{bmatrix}.$$

All three steps were replacements, so $\det A_t$ is the product of the diagonal:

$$\det A_t = t \cdot t \cdot t \cdot \left(t - \frac{1}{t^3}\right) = t^4 - 1 = (t - 1)(t + 1)(t^2 + 1).$$

Completeness. The elimination assumed $t \neq 0$, and $t = 0$ is checked separately: at $t = 0$ the matrix is the permutation matrix of the cycle $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$, which has $\det = -1 \neq 0$. Since $t^2 + 1 > 0$ for every real t , the factors $t \pm 1$ give the only real zeros. (Over $\mathbb{C}$ there would also be $t = \pm i$; the problem asked for real values.)

$\det A_t = (t - 1)(t + 1)(t^2 + 1)$, so $\det A_t = 0$ exactly at $t = 1$ and $t = -1$.

Problem 8

Compute $\det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 4 \end{bmatrix}$ by row echelon form. Fractions are unavoidable here; carry them exactly rather than rounding.

Clear column 1 below the pivot:

$$R_2 \leftarrow R_2 - \frac{1}{2}R_1 : \quad [1 \ 3 \ 1] - [1 \ \frac{1}{2} \ 0] = [0 \ \frac{5}{2} \ 1], \quad \begin{bmatrix} 2 & 1 & 0 \\ 0 & \frac{5}{2} & 1 \\ 0 & 1 & 4 \end{bmatrix}.$$

Clear column 2 below its pivot, using the exact ratio $\frac{1}{5/2} = \frac{2}{5}$:

$$R_3 \leftarrow R_3 - \frac{2}{5}R_2 : \quad [0 \ 1 \ 4] - [0 \ 1 \ \frac{2}{5}] = [0 \ 0 \ \frac{18}{5}].$$

All steps were replacements, so $\det A$ is the diagonal product:

$$\det A = 2 \cdot \frac{5}{2} \cdot \frac{18}{5} = 2 \cdot 3 = 18.$$

$\det A = 18$

Problem 9

Compute the determinant of $A = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 2 & 4 & 1 \\ 1 & 4 & 4 & 5 \end{bmatrix}$ by reducing to row echelon form. Find a reduction that needs neither swaps nor scalings, and say why such a reduction exists here at all.

Such a reduction exists because the first pivot $a_{11} = 1$ divides every entry below it in column 1 — so clearing that column needs only integer multiples, and no fractions are ever introduced. The same luck repeats at each later pivot.

Column 1:

$$R_2 \leftarrow R_2 - R_1, \quad R_4 \leftarrow R_4 - R_1 : \quad \begin{bmatrix} 1 & 4 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 2 & 4 & 1 \\ 0 & 0 & 3 & 5 \end{bmatrix}$$

Column 2 (pivot -2 , and $2 = (-1) \cdot (-2)$, so this is again a replacement):

$$R_3 \leftarrow R_3 + R_2 : \begin{bmatrix} 1 & 4 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 3 & 5 \end{bmatrix}$$

Column 3:

$$R_4 \leftarrow R_4 - R_3 : \begin{bmatrix} 1 & 4 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

Every step was a replacement, so the determinant is the diagonal product $1 \cdot (-2) \cdot 3 \cdot 4 = -24$.

$\det A = -24$

Problem 10

Let $B_t = \begin{bmatrix} 1 & 1 & t \\ 0 & 1 & 1 \\ 1 & 0 & t \end{bmatrix}$. Show by row reduction that $\det B_t = 1$ for *every* t . Then explain why expanding along the first row would have made this much harder to see.

Two replacements, and t disappears completely:

$$R_3 \leftarrow R_3 - R_1 : \begin{bmatrix} 1 & 1 & t \\ 0 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix}, \quad R_3 \leftarrow R_3 + R_2 : \begin{bmatrix} 1 & 1 & t \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

The echelon form is upper triangular with diagonal 1, 1, 1, and both steps were replacements, so

$$\det B_t = 1 \cdot 1 \cdot 1 = 1 \quad \text{for all } t.$$

Why expansion hides this: along the first row you get $1 \cdot (\text{cofactor}) - 1 \cdot (\text{cofactor}) + t \cdot (\text{cofactor})$, and each cofactor is a 2×2 determinant still containing t . The t -terms must then be expanded and cancelled by hand — four 2×2 determinants, with a real chance of a sign slip, to rediscover what two row operations showed at a glance. The parameter t is not a problem for row reduction; it only makes the bookkeeping in expansion longer.

$\det B_t = 1$ **for every** t .

Problem 11

Let A be $n \times n$. Prove that $\det A \neq 0$ if and only if the row echelon form of A has a pivot in every row. (Your proof may use the three determinant effects of elementary row operations, but it must not use the general theory of rank.)

Write E for the row echelon form of A , and let s be the number of swaps and $c_1, \dots, c_k$ the scaling factors used on individual rows on the way to E . Every other operation is a replacement, which leaves the determinant alone. Hence

$$\det E = (-1)^s c_1 \cdots c_k \cdot \det A.$$

The factor $(-1)^s c_1 \cdots c_k$ is a single nonzero number, so $\det E = 0$ **if and only if** $\det A = 0$.

Now analyze E . It is upper triangular, so $\det E$ is the product of its diagonal entries — and the diagonal entries of an echelon matrix are exactly its pivots.

($\Rightarrow$) If E has a pivot in every row, all n diagonal entries are nonzero, so $\det E \neq 0$, hence $\det A \neq 0$.

($\Leftarrow$) If E is missing a pivot in some row, that row is entirely zero, so $\det E = 0$, hence $\det A = 0$.

Therefore $\det A \neq 0$ exactly when every row of the echelon form carries a pivot. $\square$

Problem 12

True or false? Justify each answer (if false, give a counterexample or correct the statement).

(a) True. A zero row makes the matrix singular, so its determinant is 0; and a zero row of $\text{REF}(A)$ survives back through replacements (which cannot create or destroy a zero row), while swaps and scalings multiply the determinant by a nonzero factor. Equivalently, use Problem 11.

(b) False. Row replacements leave the determinant *unchanged* — that is exactly what makes the method work. The correct statement: $\det(\text{REF}(A)) = \det A$ provided the reduction used only replacements; a swap or a scaling introduces a factor of -1 or c that must be tracked.

(c) True. Replacements never change the determinant, so if the result has determinant 0 then so did A .

(d) True. Triangularity alone is enough; echelon form is stronger than needed. For an upper triangular matrix the determinant is the product of the diagonal entries, and induction on the size proves it by expanding along the last row.

(e) True. If the first row of an echelon matrix were zero, then every row below it is also zero (zeros below the diagonal, and a zero first row means no pivot has been placed), so the determinant would be 0 — contradicting the hypothesis. In an echelon matrix, a zero row can only appear at the *bottom*.